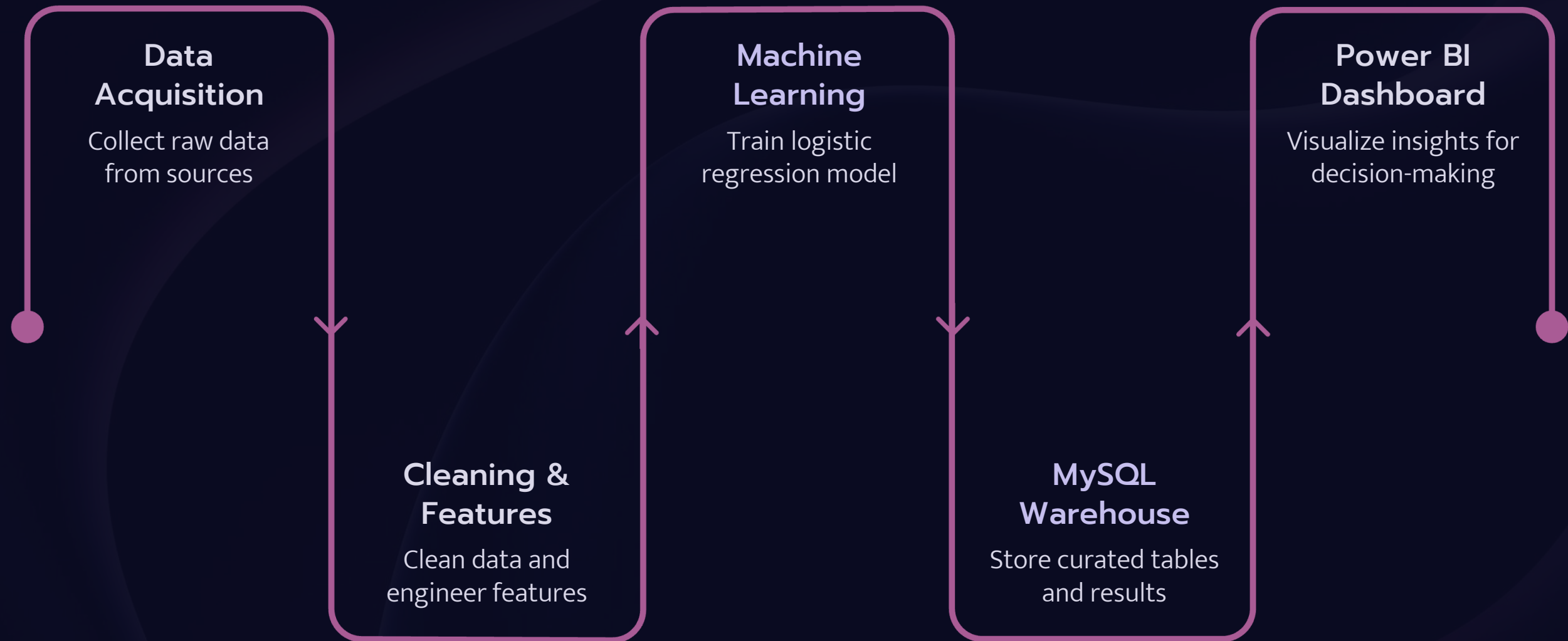




End-to-End Loan Default Analysis & Portfolio Risk Assessment

A comprehensive data pipeline translating complex machine learning outputs into actionable business intelligence — from raw data to executive dashboards.

Project Architecture & Lifecycle



By migrating data from raw Python scripts through structured SQL queries into Power BI, the pipeline establishes a single source of truth for portfolio risk management.

2,010

TOTAL RECORDS
ANALYZED [CITE: 2]

29

DATA FEATURES
EVALUATED [CITE: 2]

60.9%

BASELINE DEFAULT RATE
[CITE: 2]

Data Quality & Portfolio Snapshot

2,010

Total Records Analyzed

29

Data Features Evaluated

60.9%

Baseline Default Rate

Data Cleaning & Preprocessing



Key Preprocessing Actions

Missing Value Resolution

Fixed 81 missing education, 140 employment status, and 107 credit score records.

Duplicate Removal

Purged 6 fully duplicated rows to ensure statistical integrity.

Data Standardization

Standardized DTI ratios and handled missing interest rate values.

Exploratory Data Analysis Highlights

⚠ The 60.9% raw default rate signals a **high-risk portfolio** — traditional metrics like DTI and Credit Score remain the most reliable heuristics.



Credit History

Strong inverse relationship between **credit score** and likelihood of default.



Late Payments

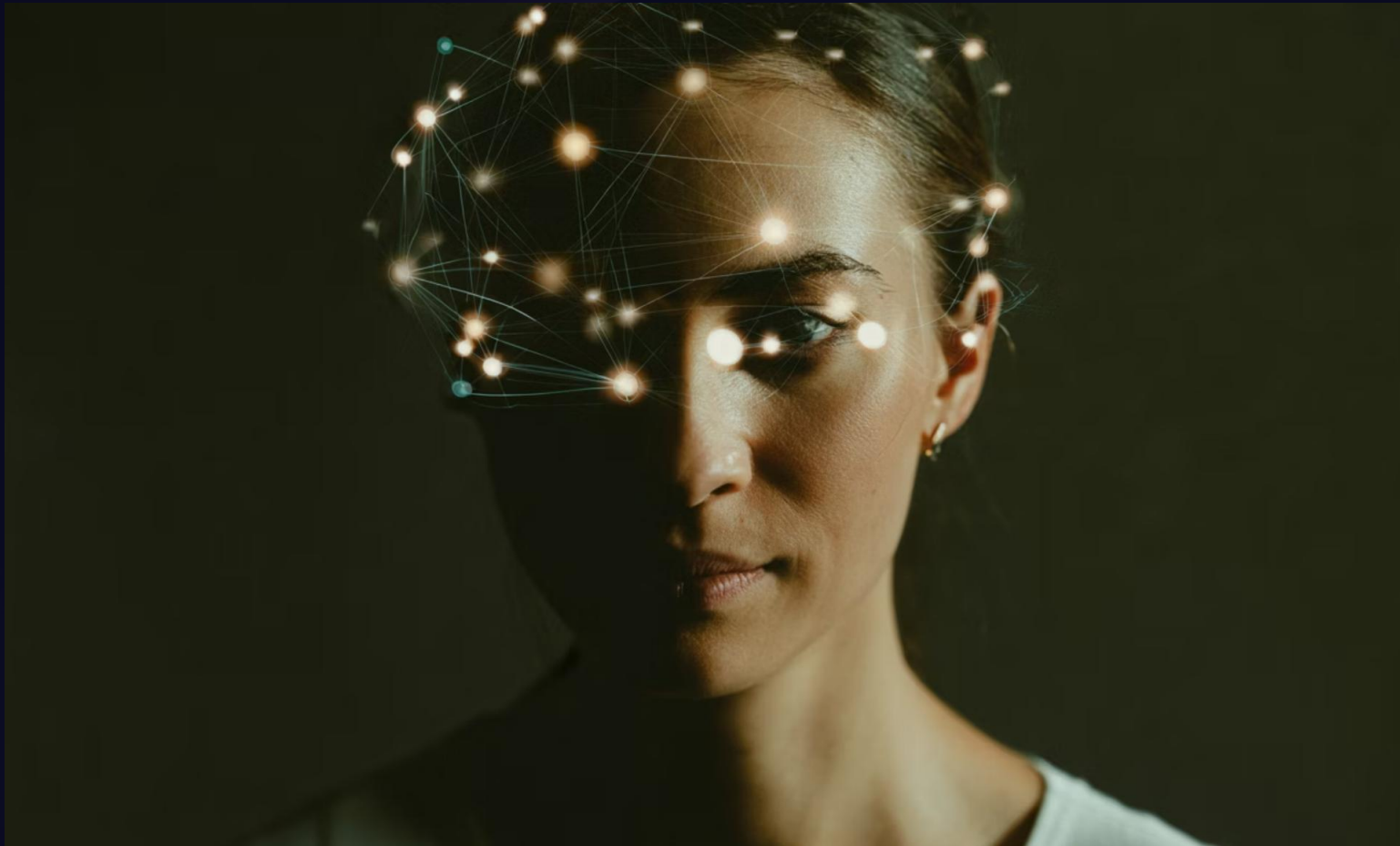
num_late_payments emerged as the primary leading indicator of future default risk.



Income Stability

Annual income and months employed gauge long-term payment capacity.

Predictive Modeling & Machine Learning



Model Pipeline

01

Algorithm

Logistic Regression for binary classification
(Default vs. No Default).

02

Feature Scaling

StandardScaler normalizes variables;
SimpleImputer handles null values.

03

Evaluation

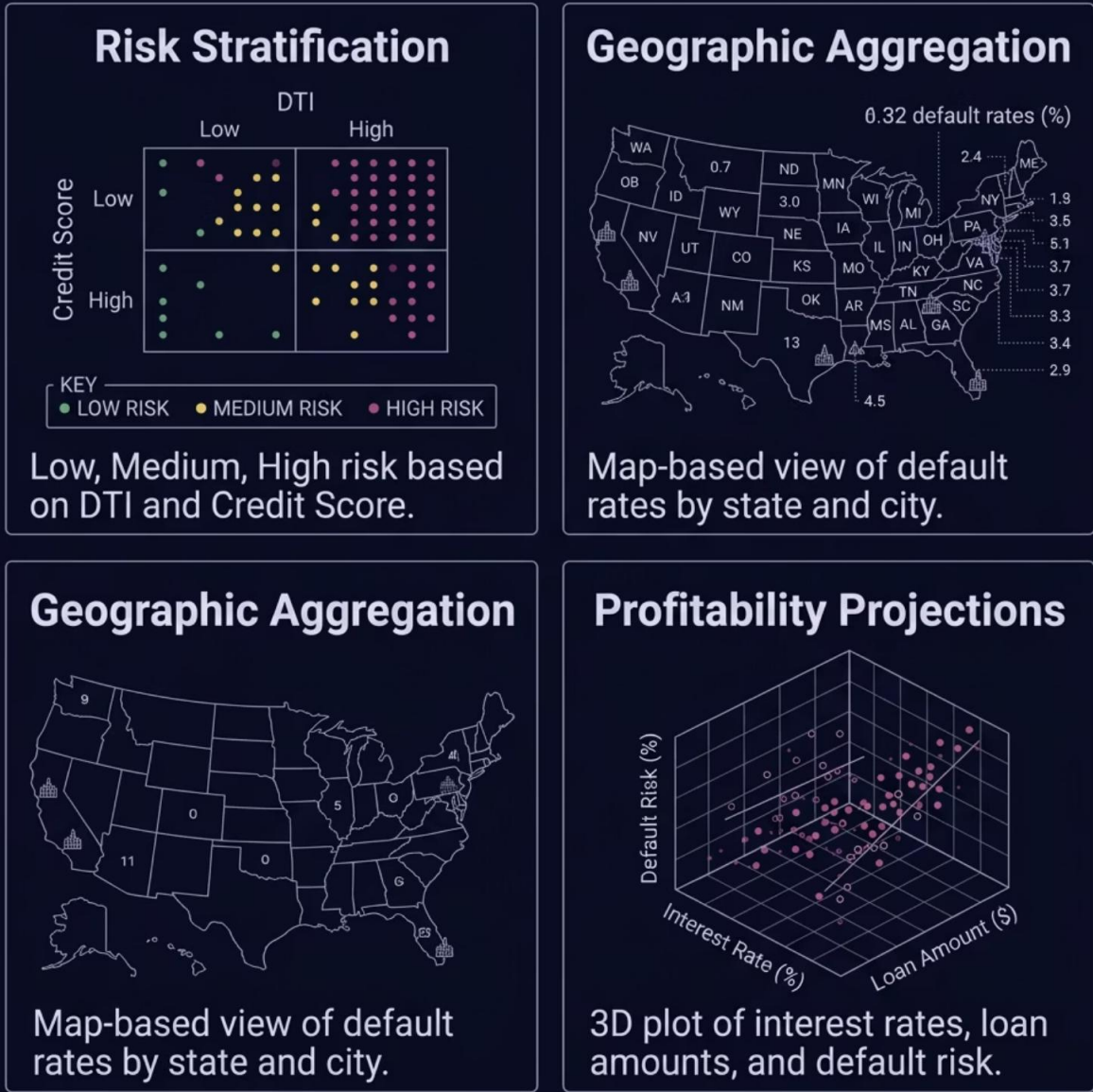
Verified via Classification Reports and ROC-AUC
scoring.

Output: *loan_default_clean_analysed.csv* — the
cornerstone for all BI operations.

Business Analysis via MySQL

Why a SQL Layer?

Raw Python models are inaccessible to business teams. MySQL bridges data science and operational logic — pre-aggregating data for fast, stable Power BI queries.



Power BI Dashboard: View 1 – Executive KPIs

Portfolio Health Cards

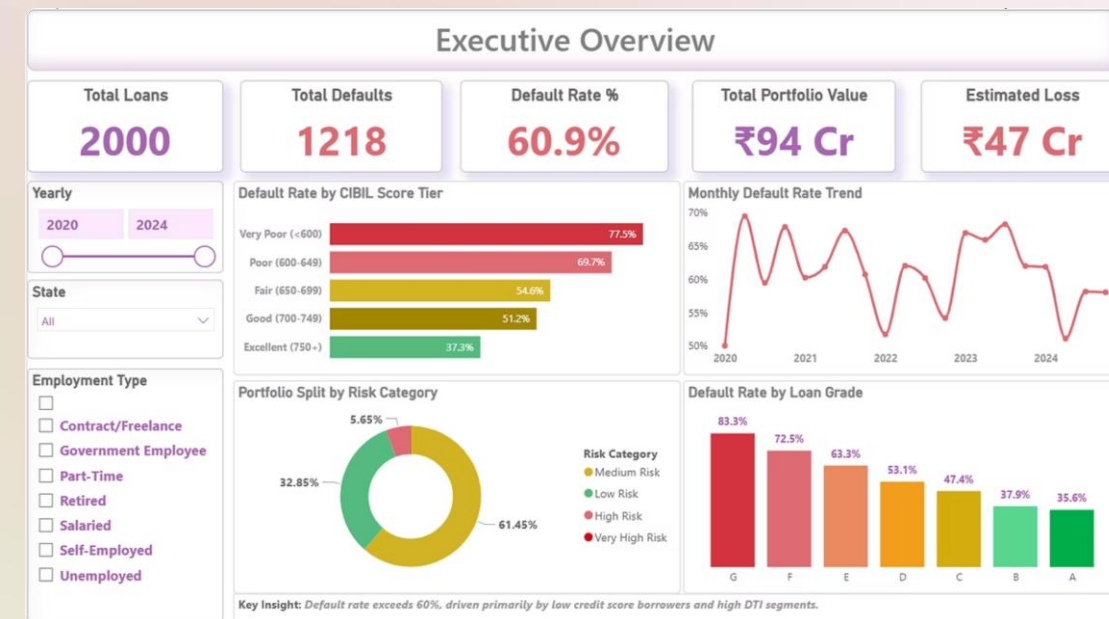
Total Loan Volume, Average Loan Amount, and the ~60.9% Critical Default Rate.

Trend Lines

Temporal view of application dates vs. default occurrences by quarter.

Demographic Split

Donut charts breaking down portfolio by gender and broad age categories.



Power BI: Risk Deep Dive & Demographic View

View 2 — Risk & Default

For Risk Management & Underwriting teams.

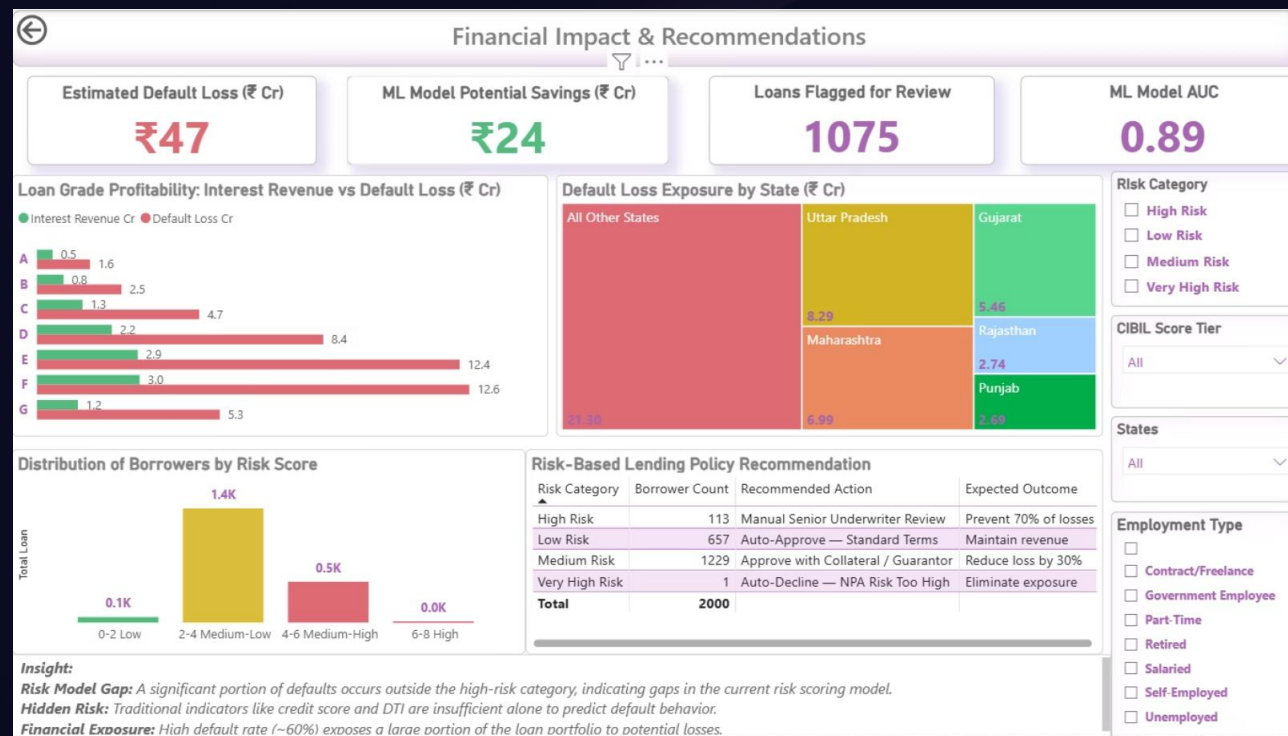
- **Credit Score Risk Bins:** Defaults vs. performing loans across score brackets.
- **DTI Scatter Plot:** Default density at higher Debt-to-Income ranges.
- **Late Payment Matrix:** Heatmap correlating late payments with default probability.



View 3 — Demographic & Operational

For Operations & Marketing executives.

- **Geographic Heatmap:** Default rates across Indian states (Gujarat, Maharashtra) and cities (Surat, Nagpur).
- **Employment Status:** Default rates among Salaried, Freelance, Part-Time, and Retired borrowers.
- **Education vs. Risk:** Educational attainment correlated with financial stability.



Strategic Recommendations

1

Tighten Underwriting

Given the 60.9% default rate, impose hard caps on acceptable DTI ratios immediately.

2

Deploy ML in Real-Time

Integrate the Logistic Regression model into the live application pipeline to flag high-risk applicants proactively.

3

Regional Risk Policies

Adjust interest rates and promotional offers based on geographic risk profiles from Dashboard View 3.

4

Weekly Dashboard Review

Risk Committee reviews Power BI dashboards weekly to catch emerging trends before they impact quarterly earnings.

✓ This pipeline shifts our posture from **reactive damage control** to **proactive risk management**.